03-621 Week 8

Advanced Quantative Genetics

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Complex and Quantitative Traits

Importance of Complex Trait Analysis in Humans

1. Identification of genetic variations that act as disease risk factors:
   (a) Basis of tests to determine predisposition or susceptibility, allowing early diagnosis and thus more successful treatment
   (b) The identified genes (or their networks) may serve as targets for therapies
   (c) Precision Medicine: understanding the genetic heterogeneity of a disorder may allow treatments better tailored to the individual
2. Defining the genetic variables simplifies analysis of the environmental risk factors.

Essential Features of Complex Traits

The phenotypes can be described quantitatively. Each phenotype of a complex trait can be assigned:

- a quantitative value x
- a frequency y in a population

Each trait exhibits a **distribution** of "phenotypic values" in a population

Three Types of Complex Traits

Continuous Traits

- Vary continuously from one phenotypic extreme to the other
- Cumulative small effects by many genes, plus environment
- Example: height, weight, hand size, skin color, etc.

Meristic Traits

- Phenotype corresponds to one of several **discrete** categories
- Additive effects of alleles at multiple genes
- Example: Number of scales on a fish, number of offspring at a time

Threshold Traits

- Only two phenotypic classes ("dichotomous")
- Each individual has an underlying **total liability** ("risk") to express the trait.
- Individuals whose total risk (genes + environment) exceeds a threshold value will express the trait.
- Example: whether or not an individual has schizophrenia

Complex Traits exhibit "Non-Mendelian" Inheritance

- The same genotype can yield different phenotypes (depending environments)
- Many different genotypes can yield the same phenotypes
- They cannoy be followed easily through pedigrees
- It can be difficult to determine that they are heritable at all.

If inheritance is particulate (per Mendel), why do we get this behavior for most traits?

- Continuous phenotype distributions
- and "regression towards the mean": on average, parents of different heights have children of intermediate height

Distributions of Complex Trait phenotypes can usually be approximated by a Gaussian Distribution (aka Normal)

- Populations can differ in Mean Phenotypic Value and Phentypic Variance for any given complex trait

Complex Traits Summary

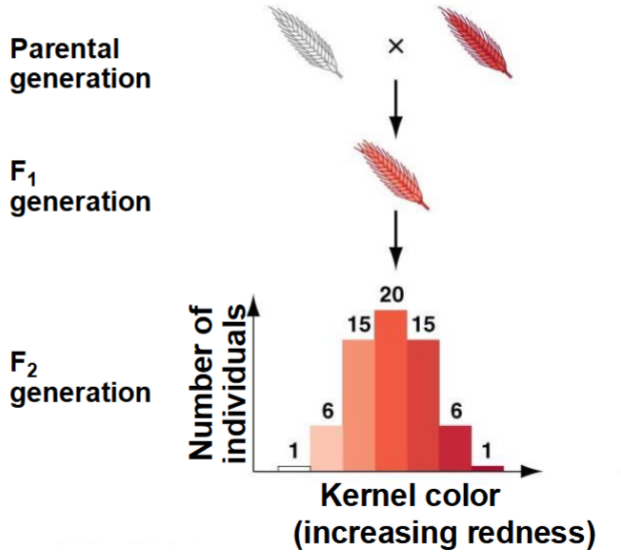
- What are complex traits?
  - Multiple genetic and environmental factors
  - Complex inheritance
- Three types of complex traits
  - Continuous: height, weight, blood pressure
  - Meristic: # puppies per litter
  - Threshold: schizophrenia, diabetes
- Distributions of phenotypes approximated by a Normal distribution

Quantitative Traits

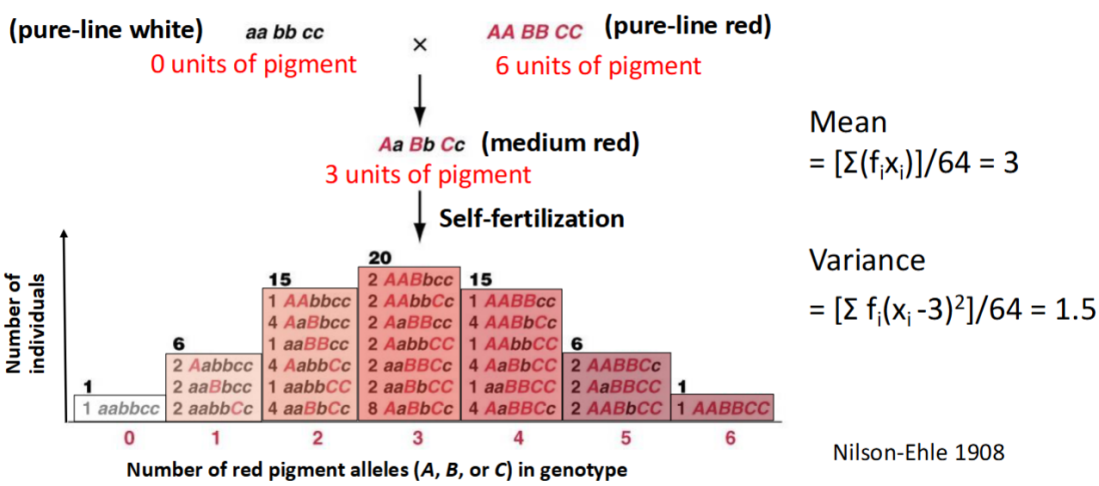
Quantitative Traits are Based on Polygenic Inheritance

Determined by the combined actions of many genes:

- Alleles at each gene contribute a small (additive) amount to the differences in phenotypic value
- Thus, the population usually exhibits a Gaussian distribution of phenotype values
- For example, wheat kernel color is a quantitative trait.



7 color phenotypes (under controlled environment)



- The distribution of genotypes corresponds to the 7 color phenotypes.
- Superimposed environmental effects can smooth out the distribution even more

Polygenic Inheritance can Give Rise to Fine-Grained Phenotypic Variation

With just two alleles per gene: # genotypes $= 3^n$.

- There are three since (homozygous dominant, homozygous recessive, and heterozygous)

Relative Importance of Genotype and Environment in Complex Traits

- Can we use phenotypic mean? Why?
- How about genetic versus environmental contributions to the variance in phenotype?

Variance due to Genotype Plus Environment

When Genotype and Environment act independently, the overall phenotypic variance is the sum of the variance due to genotype differences plus the variance due to environment differences.

$$\mathrm{Var}_p = \mathrm{Var}_g + \mathrm{Var}_e$$

where

- Var_p represents phenotypic variance
- Var_g represents genotypic contribution
- Var_e represents environmental contribution

Some examples of independence include

- serum cholesterol levels in humans (genotype + diet)
- milk production in cattle

Non-Independence: Genotype by Environment Interaction

Environmental effects on phenotype differ according to genotype.

- The G-E interaction can change the relative ranking of the genotypes with respect to phenotypic value
- For example: Maize (corn)
  - Suppose A and B are genetically distinct varieties (different inbred lines/genotypes)
  - A might perform better in a poor environment
  - B might perform better in a rich environment

Non-Independence: Genotype by Environment Association

Different genotypes are not distributed randomly with respect to the environment

- Certain genotypes are preferentially associated with certain environments, which may increase or decrease the mean phenotypic value
- For example, in the dairy husbandry: some farmers feed each cow in proportion to its level of milk production
- In horse racing: the fastest horses are assigned to the best trainers and jockeys

To what extent is the phenotype due to genetics vs environment?

‘’‘Nature vs Nurture”

- **Relative risk (sibling):**

$$\lambda_s = \frac{\text{\% affected among siblings of affected}}{\text{\% affected in general population}}$$

  - λ_s ranges from 1 to near infinity.

- **Twin Concordance:**
  - % of twin pairs where both are affected if one is affected.

$$C = \frac{AA}{AA + AN + NA}$$

  - Monozygotic: ranges from near 0 (if raised apart) to 1.0
  - Dizygotic: ranges from near 0 (if raised apart) to 0.5

- How does a shared environment impact λ_s and C values?
- Relationship to gene number is not clear.

Examples

Disorder	Relative risk for sib (λ_s)	Twin Concordance (%) MZ Twins*	DZ Twins*
Type 1 diabetes	15.0	42.9	7.4
Type 2 diabetes	3.0	34.0	16.0
Crohn's disease	25.0	37.0	10.0
Multiple sclerosis	20.0	25.3	5.4
Cleft lip	40.0	35.0	5.0
Breast cancer, female	2.0	6.5	5.5
Peptic ulcer		64.0	44.0
Tuberculosis		51.0	22.0
Rheumatoid arthritis		50.0	8.0
Parkinson disease		15.5	11.1
Ulcerative colitis		7.0	3.0
Autism spectrum disorder	6.5	60.0	7.0
Schizophrenia	9.0	40.8	5.3
Alzheimer disease (late)	4.0	32.2	8.7
Dyslexia		64.0	40.0
Major affective disorder		60.0	20.0
Alcoholism		40.0	20.0

Genetic vs Environmental Contributions to Phenotypic Variance

$$\sigma_p^2 = \sigma_g^2 + \sigma_e^2 + \sigma_{gxe}^2$$

- σ_g is the independent effect of genotype
- σ_e is the independent effect of environment
- σ_{gxe} is the effect of gene and environment interaction

So how much of the variance is due to genotype, and how much is due to environment?

Importance of the Relative Magnitude of σ_g^2

- Determines how much the phenotype can be altered by:
  - natural selection (evolution)
  - artificial selection (animal / plant breeding)
- For **human** complex disease traits:
  - Determines to what extent risk can be reduced by modifying environmental factors
    - diet, exercise, stress, exposure to risk factors, ...
    - (versus using drug therapies with side effects)
  - Determines how difficult it will be to identify the genes that influence susceptibility / risk
    - harder if <u>many</u> genes make small contributions each, or if environmental effects are large

Estimation of Variance Contributions

We have to separate the effects of Genotype and Environment. We can use genetics to reduce or eliminate Genotypic variance.

To do reduce or eliminate genotypic variance, we first inbreed, to obtain two lines of pure-breeding organisms, then cross them.

- Recall that inbreeding reduces heterozygosity. (e.g., enough times would create pure-breeding lines)
- Cross the pure-breeding lines, so all the F1 progeny is heterozygous in mostly everything.

For a genetically **heterogeneous** population, our formula is

$$V_p = V_g + V_e$$

For a genetically **uniform** population, it is

$$V_p = 0 + V_e$$

Therefore, if we have two phenotypically divergent inbred lines (P_1 and P_2), and we cross them, we get a genetically uniform population at F1.

$$V_{pF1} = 0 + V_e = V_e$$

Additionally, note that the parental generation is also genetically uniform, as in:

$$V_{pP1} = 0 + V_e = V_e \qquad V_{pP2} = 0 + V_e = V_e$$

Then, we cross the F1 generation to make a genetically heterogeneous population

$$V_{pF2} = V_g + V_e$$

At this point, both V_g and V_e can be calculated, since the environment is probably the same for all generations. If the genotype and environment are **independent**, then $V_{pP1} = V_{pP2} = V_{pF1}$.

Estimating Genetic Variance for a Human Trait

With humans, we don't have inbred lines or controlled crosses. However, for identical (i.e. monozygotic) twins, $\sigma_g^2 = 0$. Therefore, we get $\sigma_p^2 = 0 + \sigma_e^2$.

- How do we get σ_g^2 in this case?
- Genotype and environment are what make them similar, so we can cancel the effect of environment by analyzing the covariance between many pairs of identical twins raised apart in different and random environments.
- If X_i and Y_i are the members of a twin pair i:

$$COV_{X,Y} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{N-1} = COV_g + COV_e$$

- For monozygotic twins raised apart in different random environments,
  - Environment covariance $COV_e = 0$ (because random)
  - Therefore, $COV_g = COV_{X,Y}$
- However, $\sigma_g^2 = 0$ for any pair of identical twins, so: $(X_{gi} - \bar{X}_g) = (Y_{gi} - \bar{Y}_g)$.
- Therefore,

$$COV_{X,Y} = \frac{\sum(X_{gi} - \bar{X}_g)^2}{N-1} = \frac{\sum(Y_{gi} - \bar{Y}_g)^2}{N-1} = V_g$$

The genetic contribution V_g is equal to the covariance between identical twin pairs raised apart.

V_g from Dizygotic twins

We can estimate V_g using dizygotic twins instead of monozygotic twins, which might be more practical, but not as good.

With a fraternal twin (FT) pair (half alleles shared), we have

$$V_{pFT} = \frac{1}{2}V_g + V_e$$

This simplifies to

$$V_g = 2[V_{pFT} - V_{pIT}]$$

This can yield overestimates for V_g because:

1. Embryonic membranes are shared between identical twin (IT) pairs
2. Different sexes in half of FT pairs
3. GxE interaction increases FT variance but not IT.
4. Greater similarity in treatment of ITs
5. "Private" environment effects shared by pairs of twins raised together will appear to be genetic

H^2 can also be estimated by comparing identical and fraternal twins.

- We just saw that:

$$V_e = V_{pIT}$$
$$V_g = 2[V_{pFT} - VpIT]$$

  Therefore,

$$H^2 = \frac{V_g}{V_g + V_e} = \frac{2[V_{pFT} - V_{pIT}]}{2V_{pFT} - V_{pIT}}$$

Again, this can yield overestimates for V_g for the same reasons as above.

Evolutionary Impact of the Contributing Gene Number

The number of genes influences the amount by which a population can be genetically improved by selective breeding (or adapted by natural selection).

- The larger the number of genes, the greater this potential
- If the number of genes is large, selective breeding can even generate a population in which the phenotypic value of *every* individual **greatly exceeds that of the best individuals in the starting population**.

Reason 1: Genotype Frequencies

- In a real starting population of *limited size*, many of the possible genotypes are never formed because their expected frequencies are too low.
- But, as selection changes the allele frequencies, these genotypes may become more likely, allowing further directional selection in future generations.

Consider a continuous trait determined by 5 genes with allelic pairs:

$$A\ a \quad B\ b \quad C\ c \quad D\ d \quad E\ e$$

Suppose the most desireable phenotype is associated with the genotype

$$aa\ bb\ cc\ dd\ ee$$

- Before selection, the frequency of each one might be 0.1.
- After a certain degree of selection, the frequency of each one might be 0.4.
- This 4x change in allele frequencies gives a million-fold change in genotype frequency. ($0.1^5 = 1 \times 10^{-10}, 0.4^5 = 1 \times 10^{-4}$)

Reason 2: Probability of Mutations

The larger the number of genes, the greater the probability that new mutations will arise in the population that further increase the phenotypic range.

- (Because the target size for such mutations is larger)

If $\mu = 10^{-6}$ mutations per gene per generation:

- A 3 gene trait would have 3×10^{-6} new alleles per generation that may affect the trait.
- A 20 gene trait would have 2×10^{-5}

Impact of the Contributing Gene Number for Human Disease Traits

- Determines how difficult it will be to identify the genes that influences susceptibility / risk
- Harder if alleles at many genes each make small contributions

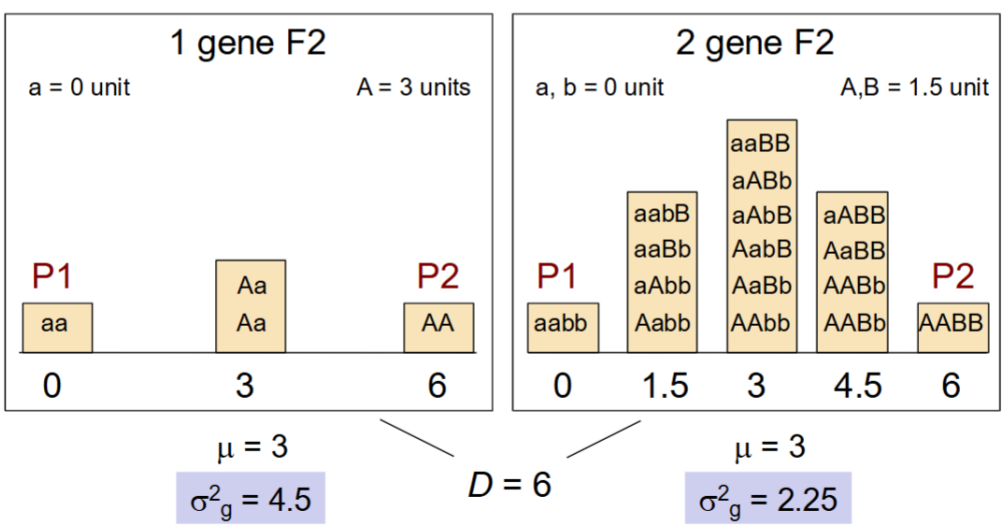
The number of genes cannot be deduced from phenotype distributions alone.

This is because we can approximate the same normal distribution with different numbers of genes, adjusting the number of alleles, allele contributions, and allele frequencies.

- However, there is a way we can estimate the gene number using crosses.

Relation of V_g to Gene Number

For a given difference D in phenotypic means between P_1 and P_2, the genotypic variance V_g in the F2 decreases with the number of contributing genes.



$$V_g = \frac{D^2}{8n}$$

- D = difference between the **means** of the original inbred parent strains P_1 and P_2

- n = gene number
- V_g = variance

This relationship assumes:

- Random mating
- Additive effects by alleles and genes. (No dominance, no epistasis)
- Equal contribution by all genes
- Genes are unlinked
- Same environment

Proper Interpretation of Broad-Sense Heritability

$$H^2 = \frac{\sigma_g^2}{\sigma_p^2}$$

- H^2 is important for estimating the genetic contribution to *variance* in quantitative traits, but it can be misinterpreted.
  - Applies to populations, not individuals
  - NOT the proportion of the phenotypic value that is due to genotype
  - NOT fixed: varies with environment and genetic composition
  - Cannot be used to predict progeny phenotype
- Reason: Only the **additive** genetic effects lead to heritable differences in phenotypic mean, but

  $$V_g = V_A + V_D + V_I$$

  (**A**dditive + **D**ominance + **E**pistasis effects)

A Predictive Measure of Heritability: Narrow-Sense Heritability h^2

Defined as: Ratio of **additive** genotypic variance to total phenotypic variance

$$h^2 = \frac{V_A}{V_p} = \frac{V_A}{V_g + V_e}$$

- V_A is the genotypic variance due to **additive effects of alleles and genes**

  $$V_g = V_A + V_D + V_I$$

h^2 is used for predicting phenotypic **means** of a population after group selection, or among progeny of individuals

- How to obtain h^2
  - Parent-Offspring Regression
  - Selection experiments: the "Breeder's Equation"
  - Incidence distributions for Threshold Traits